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1. What is Python?

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=> Python is a high-level, general-purpose programming language.

=> It is widely used in IT industry because it is

- Simple
- Powerful
- Easy to learn
- Beginner friendly

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2. Why Python so Popular?

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=> Python is popular because its syntax is very close to English.

```
print("Hello World");
```

=> This program is very easy to understand even for beginners.

In many programming languages, we need to write more code to display one message.

=> But in Python, we can display output with one simple line.

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3. History of Python

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=> Python was created by Guido van Rossum.

=> Python development started in the late 1980s.

=> Python was first released in 1991.

=> Python was designed to be simple, readable, and powerful.

1980s : Python development started

1991 : Python First Released

2000 : Python 2 Released

2008 : Python 3 Released

Today : Python is widely used in web, AI, data science, automation, DevOps, and backend development

Python has two major historical versions:

Python 2
Python 3

Python 2 is old and no longer recommended for new projects.

Python 3 is the current and recommended version.

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Features of Python
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- 1) Simple and easy to learn
- 2) Free and Open source
- 3) High Level Language
- 4) Platform Independent
- 5) Dynamically Typed
- 6) Object Oriented (class, object)
- 7) Powerful Third Party Libraries

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Python Usecases
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- 1) Web Development
- 2) Data Science
- 3) Machine Learning
- 4) Artificial Intelligence
- 5) Automation
- 6) DevOps Practices
- 7) Cloud Computing
- 8) Cyber Security
- 9) Gaming
- 10) Web Scaraping
- 11) API Development

Python in Web Development and Microservices

Python we can use to develop web applications and APIs.

~~~~~ Popular Frameworks : Django, Flask, FAST API

## Python in Data Science, AI & Machine Learning

=> Python is very popular in data science because of its libraries suport

~~~~~ Python Libraries : Numpy, Pandas, Matplotlib, seaborn, Scipy...

scikit-learn, Tensorflow, PyTorch,

Keras....

Python in Automation

=> Python is very useful for automation.

- Rename multiple files
- Send Emails Automatically
- Generate Reports

Python in DevOps and Cloud

=> Python is used in DevOps for automation and scripting.

- Servers Automation
- Log Analysis
- AWS Automation
- Docker Scripts
- K8S Automation
- CI CD Scripts

~~~~~ Python Libraries : boto3, fabric, requests

# Python in Cybersecurity

Python is also used in cybersecurity.

- Network Scanning
- Security Automation
- Malware Analysis
- Password strength checking
- Vulnerability testing tools

~~~~~ Python Libraries : scapy, cryptography, requests

Python in Web Scraping

=> Web scraping means extracting data from websites.

~~~~~ Python Libraries : BeautifulSoup, scrapy

# Python in Testing

=> Python is used for automation testing.

~~~~~ Python Libraries : pytest, unittest, selenium

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Python Setup
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1. Download Python

Go to the official Python website: <https://www.python.org/downloads/>

Download the latest Python version for your operating system.

2. Install Python

Run the downloaded installer.

Very important: Select this option before installing:

Add Python to PATH Then click: Install Now

3. Check Python Installation

Open Command Prompt or Terminal and type:

```
python --version or py --version
```

You should see output like: Python 3.x.x

4. Install Code Editor (IDE)

Install any one code editor: "PyCharm" or "Visual Studio Code"

For beginners, PyCharm is easy to use for Python practice.

<https://www.jetbrains.com/pycharm/download/?section=windows>

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Python Program Execution Flow
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1. Write Python Code

We write Python code in a file with .py extension.

Example: `print("Hello Python")`

File name: demo.py

2. Run the python code using below command

syntax : `python demo.py`

3. Python Interpreter Reads Code

Python uses an interpreter to read the program.

Interpreter means: It reads and executes code line by line

4. Source Code Converts into Byte Code

Python source code is converted into byte code.

Source Code → Byte Code

Source code means the code written by programmer.

Byte code is an intermediate code understood by Python Virtual Machine.

5. Byte Code Goes to PVM

PVM means Python Virtual Machine. PVM executes the byte code.

Byte Code â†’ Python Virtual Machine -> Output to use